

ECO7101
CHAPTER-23
MONOPOLY MARKET

Monopoly is at the opposite end of the market structure spectrum from perfect competition. Much of this chapter focuses on the differences between the perfectly competitive firm and the monopoly firm.

Monopoly

A theory of market structure based on three assumptions: There is one seller, it sells a product that has no close substitutes, and the barriers to entry are extremely high.

Properties

1. There is one seller. In effect, the firm is the industry. Contrast this situation with perfect competition, where many firms make up the industry.
2. The single seller sells a product that has no close substitutes. Because there are no close substitutes for its product, the single seller—the monopolist or monopoly firm—faces little, if any, competition.
3. The barriers to entry are extremely high. In the theory of perfect competition, a firm can enter the industry easily. In the theory of monopoly, entering the industry is very hard (if not impossible). Extremely high barriers keep out new firms.

Examples of monopoly include many public utilities (local public utilities such as electricity, water, and gas companies) and the U.S. Postal Service (in the delivery of first-class mail).

Barriers to Entry: A Key to Understanding Monopoly

Legal barriers, economies of scale, or one firm's exclusive ownership of a scarce resource may make it difficult or impossible for new firms to enter the market. Legal barriers include public franchises, patents, and government licenses.

A public franchise is a right that government grants to a firm and that permits the firm to provide a particular good or service and excludes all others from doing so (thus eliminating potential competition by law). For example, the U.S. Postal Service has been granted the exclusive franchise to deliver first-class mail. Many public utilities operate

under state and local franchises, as do food and gas suppliers along many state turnpikes.

In the United States, patents are granted to inventors of a product or process for a period of 20 years. During that time, the patent holder is shielded from competitors: No one else can legally produce and sell the patented product or process. The rationale behind patents is that they encourage innovation in an economy. Few people will waste their time and money trying to invent a new product if their competitors can immediately copy and sell it.

Entry into some industries and occupations requires a government-granted license. For example, radio and television stations cannot operate without a license from the Federal Communications Commission (FCC). In most states, a person needs to be licensed to join the ranks of physicians, dentists, architects, nurses, embalmers, barbers, veterinarians, and lawyers, among others.

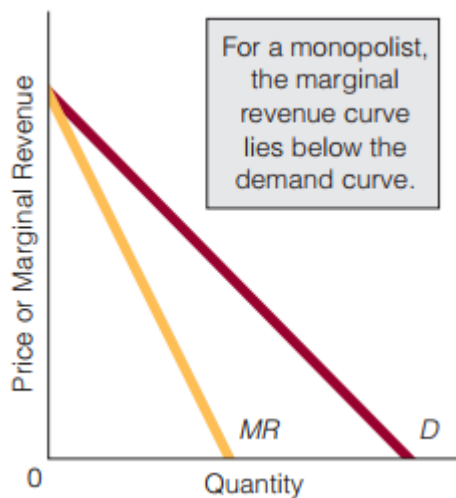
Economies of Scale In some industries, low average total costs (low unit costs) are obtained only through large-scale production. Thus, if new entrants are to compete in the industry, they must enter it on a large scale. But having to produce on this scale is risky and costly, and it therefore acts as a barrier to entry. If economies of scale are so pronounced that only one firm can survive in the industry, the firm is called a natural monopoly. Often-cited examples of natural monopoly include public utilities that provide gas, water, and electricity. A later chapter discusses government regulation of a natural monopoly.

Exclusive Ownership of a Necessary Resource Existing firms may be protected from the entry of new firms by the exclusive or nearly exclusive ownership of a resource needed to enter the industry. The classic example is the Aluminum Company of America (Alcoa), which, for a time, controlled almost all the sources of bauxite in the United States. Alcoa was the sole producer of aluminum in the country from the late nineteenth century until the 1940s. In a similar vein, many people today view the De Beers Company of South Africa as a monopoly because it controls such a large percentage of diamond production and sales. acts much like a monopolist.

The Monopolist's Demand Curve and Marginal Revenue Curve Are Not the Same

In perfect competition, the firm's demand curve is the same as its marginal revenue curve. In monopoly, the firm's demand curve is not the same as its marginal revenue curve but rather lies above its marginal revenue curve.

The relationship between a monopolist's demand curve and marginal revenue curve is illustrated in Exhibit 2. The demand curve (D) plots price and quantity; the marginal revenue curve (MR) plots marginal revenue and quantity. Because price is greater than marginal revenue for a monopolist, its demand curve necessarily lies above its marginal revenue curve.

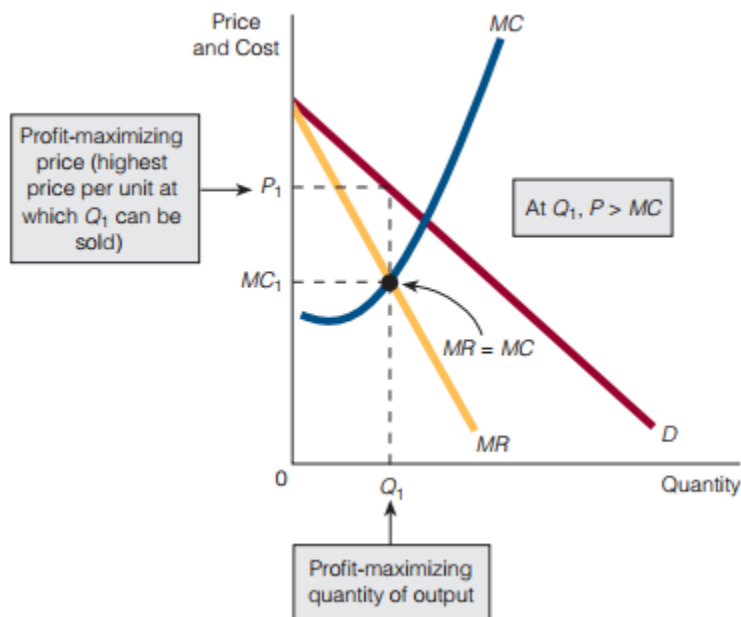


The demand curve plots price against quantity. The marginal revenue curve plots marginal revenue against quantity. For a monopolist, $P > MR$, so the marginal revenue curve must lie below the demand curve.

MONOPOLY PRICING AND OUTPUT DECISIONS

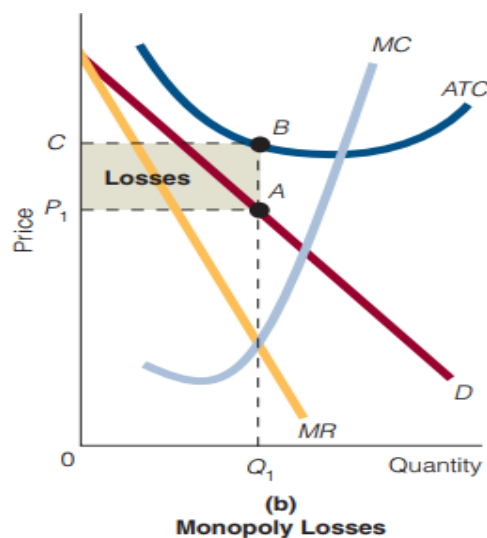
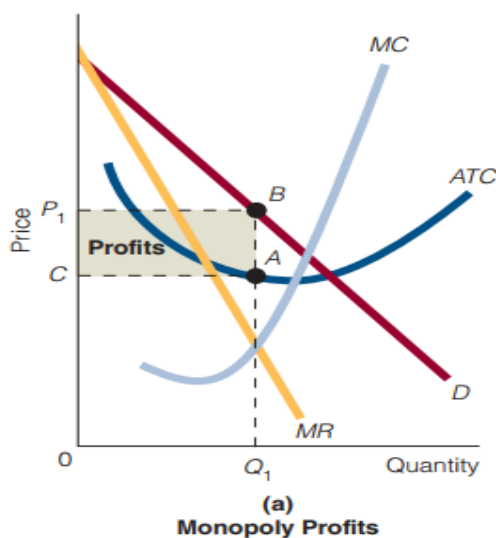
A monopolist is a price searcher—that is, a seller with the ability to control, to some degree, the price of the product it sells. A price searcher can raise its price and still sell its product. In the theory of monopoly, the monopoly firm is the industry, and the industry is the monopoly firm; they are one and the same. Thus, the demand curve for the monopoly firm is the market demand curve, which is downward sloping. In monopoly, the firm's demand curve is not the same as its marginal revenue curve but rather lies above its marginal revenue curve.

The monopolist that seeks to maximize profit produces the quantity of output at which $MR = MC$ and charges the highest price per unit at which this quantity of output can be sold.



The monopoly equilibrium is at the intersection point of MR and MC . The monopolist produces the quantity of output, Q_1 at which $MR = MC$ and charges the highest price per unit P_1 at which this quantity of output can be sold. Q_1 is the profit maximizing output. Monopoly charges a price greater than MC (i.e., $P > MC$).

Monopoly Profits and Losses



A monopoly seller is not guaranteed any profits. In (a), price is above average total cost at Q_1 , the quantity of output at which $MR = MC$. Therefore, TR (the area OP_1BQ_1) is greater than TC (the area $OCAQ_1$), and profits equal the area CP_1BA . In (b), price is below

average total cost at Q_1 . Therefore, TR (the area OP_1AQ_1) is less than TC (the area $OCBQ_1$), and losses equal the area P_1CBA .

Whether profits are earned depends on whether P_1 is greater or less than average total cost at Q_1 . In short, the profit-maximizing price may be the loss-minimizing price. Monopoly profits and monopoly losses are in the above figures. Some people argue that suggesting that a monopolist can take a loss is unrealistic. If the monopolist is the only seller in the industry, they maintain, it is guaranteed a profit. But even when a firm is the only seller of a product, it may not earn a profit. A monopolist cannot charge any price it wants for its good—only the highest price that the demand curve allows it to charge. In some instances, the highest price may be lower than the firm's average total costs (unit costs). If so the monopolist incurs a loss, as shown in the figure.

If a Firm Maximizes Revenue, Does It Automatically Maximize Profit Too?

Profit is the difference of total revenue (TR) and total cost (TC): $\text{Profit} = \text{TR} - \text{TC}$. Because TC is the sum of total fixed costs (TFC) and total variable costs (TVC), we can rewrite our profit equation as

$$\text{Profit} = \text{TR} - (\text{TFC} + \text{TVC})$$

Maximizing profit is the same as maximizing total revenue under one condition: $\text{TVC} = 0$. When $\text{TVC} = 0$, TVC falls out of our profit equation and we are left with

$$\text{Profit} = \text{TR} - \text{TFC}$$

Because TFC is constant as output increases, a rise in TR will automatically increase profit by the same amount. To illustrate, if $\text{TR} = \$100$ and $\text{TFC} = \$40$, then profit is $\$60$. If TR rises by $\$10$, to $\$110$, and TFC remains constant at $\$40$, then profit rises to $\$70$.

The rise in TR is equal to the rise in profit: $\$10$. Therefore, to maximize total revenue is to maximize profit.

Now change things a bit. Suppose TVC is not zero. We return to our profit equation:

$$\text{Profit} = \text{TR} - (\text{TFC} + \text{TVC})$$

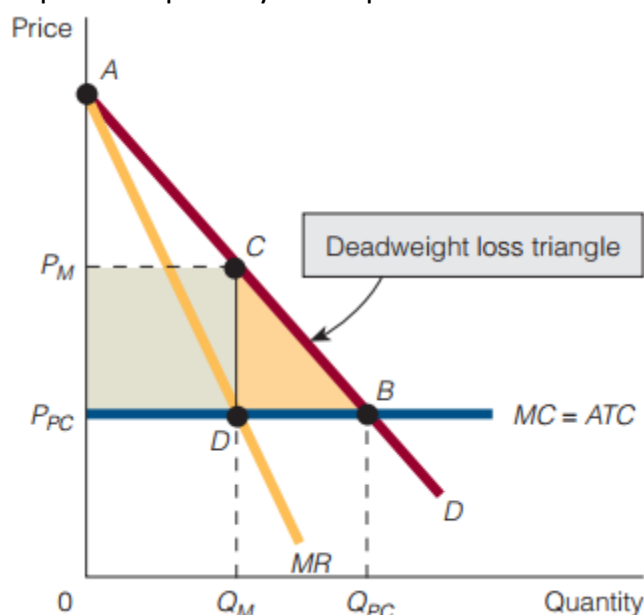
Again, let $\text{TR} = \$100$ and $\text{TFC} = \$40$. But this time $\text{TVC} = \$20$. It follows that profit is $\$40$. Now, if TR rises to $\$110$, will profit again rise by $\$10$, as it did in the previous example?

Perhaps not, because what happens to profit depends on what happens not only to TR, but to TVC too. Suppose TVC rises to \$37 as TR rises by \$110. Now profit is \$33. In other words, total revenue increased (from \$100 to \$110), but profit decreased (from \$40 to \$33).

Therefore, profit maximization is the same as revenue maximization only when there are no variable costs (i.e., when $TVC = 5 \$0$).

Deadweight Loss of Monopoly

The net value (the value to buyers over and above the costs to suppliers) of the difference between the competitive quantity of output (where $P = MC$) and the monopoly quantity of output (where $P > MC$); the loss due to not producing the competitive quantity of output.



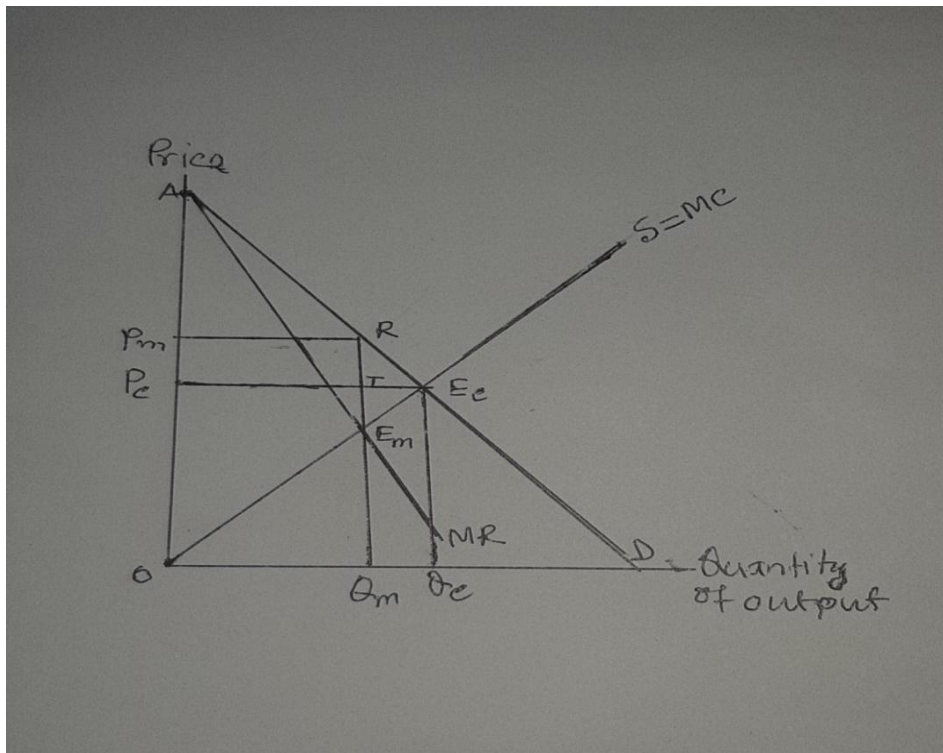
Above figure shows demand, marginal revenue, marginal cost, and average total cost curves. For simplicity's sake, assume that the product is produced under constant cost conditions, so that marginal cost equals long-run average total cost. If the product is produced under perfect competition, then output Q_{PC} is produced and is sold at price P_{PC} . At the competitive equilibrium output level, $P = MC$. If the product is produced under monopoly, output Q_M is produced and is sold at price P_M . At the monopoly equilibrium is at $P > MC$.

As the figure shows, greater output is produced under perfect competition than under monopoly. The net value of the difference in these two output levels is said to be the deadweight loss of monopoly. In the figure, the value to buyers of increasing output

from Q_M to Q_{PC} is equal to the maximum amount they would pay for this increase in output, designated by the trapezoidal area Q_MCBQ_{PC} . The costs that would have to be incurred to produce the additional output are designated by the rectangular area Q_MDBQ_{PC} . The difference between the two is the triangle DCB , the amount buyers value the additional output over and above the costs of producing the additional output. This triangular area is the loss attached to not producing the competitive quantity of output. The triangle DCB is referred to as the deadweight loss triangle.

Distinction between perfect competitive market and monopoly market

Perfect competitive market	Monopoly market
1. There are many sellers	1. There is single seller
2. Competitive firm is price taker	2. Monopoly is price setter
3. Competitive firm's demand curve is horizontal. Demand curve and MR curve are same curve.	3. Monopoly demand curve is downward sloping. MR curve lies below the demand curve
4. $P = MC$	4. $P > MC$
5. Efficient market	5. Inefficient market
6. No barriers to entry	6. There are barriers to entry



In the above diagram;

D = demand curve; S = Supply curve which is also MC curve; MR = marginal revenue curve; E_c = competitive equilibrium point where $D = S$; E_m = Monopoly equilibrium point where $MR = MC$; Q_c = competitive equilibrium output; Q_m = monopoly equilibrium output; P_c = competitive price and P_m = monopoly price.

As we see from the figure, monopoly output (Q_m) is lower than competitive output (Q_c). That means monopoly produces less than competitive firm. But monopoly sets a price, P_m which higher than competitive price, P_c .

P_mAR is the consumer surplus in monopoly market and P_cAE_c is the consumer surplus in competitive market and $P_mAR < P_cAE_c$. It implies that the monopoly reduces and extracts consumer surplus.

Due to under production by the monopolist, triangles TRE_c and E_mTE_c are lost. TRE_c is the loss of consumer surplus and E_mTE_c loss of producer surplus. These two areas neither go to producer nor to consumer. This loss is the loss of the society known as 'deadweight loss'. Since monopolist causes this loss, monopoly is inefficient.

Monopolist reduces consumer surplus by the area $P_cP_mRE_c$. P_cP_mRT goes to monopolist. Monopolist lost E_mTE_c . But $P_cP_mRT > E_mTE_c$. Monopolist here is net gainer and consumer is net loser.

Price Discrimination

A price structure in which the seller charges different prices for the product it sells and the price differences do not reflect cost differences.

The monopoly seller may sell all units of its product for the same price; that is, it may be a single price monopolist. However, under certain conditions, a monopolist could practice price discrimination, which occurs when the seller charges different prices for the product it sells and the price differences do not reflect cost differences.

Types of Price Discrimination

There are three types of price discrimination:

- Perfect price discrimination: Suppose a monopolist produces and sells 1,000 units of good X. It sells each unit separately, charging the highest price that each consumer would be willing to pay for the product rather than go without it. This practice is perfect price discrimination, sometimes called discrimination among units.
- Second-degree price discrimination: If a monopolist charges a uniform price per unit for one specific quantity, a lower price for an additional quantity, and so on, then it is

practicing second-degree price discrimination, sometimes called discrimination among quantities. For example, the monopolist might sell the first 10 units for \$10 each, the next 20 units for \$9 each, and so on.

- Third-degree price discrimination: If a monopolist charges different prices in various markets or charges different prices to various segments of the buying population, then the monopolist is practicing third-degree price discrimination, sometimes called discrimination among buyers. For example, if your local pharmacy charges senior citizens lower prices for medicine than it charges non-senior citizens, then it is practicing third-degree price discrimination

Sample questions for quiz

Define monopoly market structure

A theory of market structure based on three assumptions: There is one seller, it sells a product that has no close substitutes and the barriers to entry are extremely high.

Why does a monopolist face little or no competition?

Because there are no close substitutes for its product, the single seller—the monopolist or monopoly firm—faces little or no competition.

What is public franchise?

A public franchise is a right that government grants to a firm and that permits the firm to provide a particular good or service and excludes all others from doing so.

Why economies of scale act as a barrier to entry?

Economies of scale in some industries, low average total costs (low unit costs) are obtained only through large-scale production. Thus, if new entrants are to compete in the industry, they must enter it on a large scale. But having to produce on this scale is risky and costly, and it therefore acts as a barrier to entry.

Distinguish between legal monopoly and market monopoly.

Some economists use the term “government monopoly” to refer to a monopoly that is legally protected from competition and the term “market monopoly” to refer to a monopoly that is not legally protected from competition but because economies of scale.

Why is monopolist’s demand curve downward sloping?

Because of

- (a) Monopoly is a single seller and itself an industry.

(b) It has market power by which it can influence price.

Why does monopolist MR curve lie below the demand curve?

The demand curve plots price against quantity. The marginal revenue curve plots marginal revenue against quantity. For a monopolist, $P > MR$, so the marginal revenue curve must lie below the demand curve.

Why does a monopolist incur economic loss?

A monopolist cannot charge any price it wants for its good—only the highest price that the demand curve allows it to charge. In some instances, the highest price may be lower than the firm's average total costs (unit costs). If so the monopolist incurs a loss.

When profit maximization is the same as revenue maximization?

Only when there are no variable costs (i.e., when $TVC = 0$).

What do you mean by rent seeking?

Actions of individuals and groups that spend resources to influence public policy in the hope of redistributing (transferring) income to themselves from others is known as rent seeking.

What do you mean by deadweight loss of monopoly?

The net value (the value to buyers over and above the costs to suppliers) of the difference between the competitive quantity of output (where $P = MC$) and the monopoly quantity of output (where $P > MC$); the loss due to not producing the competitive quantity of output.

What do you mean by price discrimination?

A price structure in which the seller charges different prices for the product it sells and the price differences do not reflect cost differences.

What do you mean by first degree or perfect price discrimination?

A price structure in which the seller charges the highest price that each consumer is willing to pay for the product rather than go without it.

What do you mean by second-degree price discrimination?

A price structure in which the seller charges a uniform price per unit for one specific quantity, a lower price for an additional quantity, and so on.

What do you mean by third-degree price discrimination?

A price structure in which the seller charges different prices in different markets or charges different prices to various segments of the buying population.